FY-927

Reg. No. : .....

Name : .....

FIRST YEAR HIGHER SECONDARY
IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

Time : 2 Hours

MATHEMATICS (SCIENCE)

Cool-off time : 15 Minutes

Maximum : 60 Scores

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

Answer any 6 questions from 1 to 8. Each carries 3 scores. **(6 × 3 = 18)**

1. (a) If A and B are two set such that $A \cup B = A \cap B$, then **(1)**

(A) $A = \phi$ (B) $B = \phi$

(C) $A = B$ ✓ (D) None of these

(b) For any sets P and Q, prove that $(P \cap Q) \cup (P - Q) = P$. **(2)**

$(P \cap Q) \cup (P - Q)$
$= (P \cap Q) \cup (P \cap Q')$
$= P \cap (Q \cup Q')$
$= P \cap U = P$

2. (a) Find the value of tan 135°. **(1)**

(b) A circle of arc length 15π centimetre makes an angle $3\frac{\pi}{4}$ radian at the centre. Find the radius of the circle. **(2)**

2. (a)
$\tan 135° = -1$

2. (b)
$s = r\theta$
$15\pi = r \times 3\pi/4$
$r = 20$ cm

3. (a) Solve the inequality, $\frac{x}{3} > \frac{x}{2} + 1; x \in \mathbb{R}$. **(2)**

(b) Mark the solution obtained in part (a) to the real line $\mathbb{R}$. **(1)**

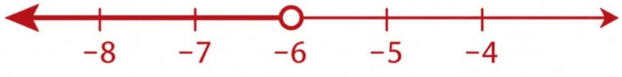
3. (a)

$x/3 > x/2 + 1$

$2x > 3x + 6$

$x < -6$

3. (b)



4. (a) Write the difference between the LCM and the HCF of 4! 5! 6!. **(1)**

(A) 24 (B) 120

(C) 696 ✓ (D) 800

(b) If ${}^{12}C_r = 495$ and ${}^{12}P_r = 11880$, then find the value of r. **(2)**

4. (a)

(C) 696

4. (b)

${}^{12}P_r / {}^{12}C_r = r!$

$11880 / 495 = 24 = 4!$

$r = 4$

5. (a) Perpendicular distance from the point (6, 7, 8) to the XY-plane is **(1)**

(A) 6 (B) 7

(C) 8 ✓ (D) 9

(b) Find the equation of the set of points which are equidistant from the points (1, 2, 3) and (3, 2, –1). **(2)**

5. (a)

(C) 8

5. (b)

Let P(x, y, z) be equidistant from (1, 2, 3) and (3, 2, −1).

$(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = (x - 3)^2 + (y - 2)^2 + (z + 1)^2$

Expanding and simplifying,

$x - 2z = 0$

6. Find the centre and radius of the circle $x^2 + y^2 - 4x - 8y - 45 = 0$. **(3)**

$x^2 + y^2 - 4x - 8y - 45 = 0$

$(x^2 - 4x) + (y^2 - 8y) = 45$

$(x - 2)^2 + (y - 4)^2 = 45 + 4 + 16$

$(x - 2)^2 + (y - 4)^2 = 65$

Centre = (2, 4), Radius = $\sqrt{65}$

7. (a) Evaluate $\lim_{x \to 0} \frac{\sin 5x}{x}$ **(1)**

(b) Evaluate $\lim_{x \to 3} \frac{x-3}{x^2-5x+6}$ **(2)**

7. (a)

$\lim_{x \to 0} (\sin 5x / x) = 5$

7. (b)

$(x - 3)/[(x - 3)(x - 2)]$

$= 1/(3 - 2) = 1$

8. (a) Find the slope of the line $x - y = 8$. **(1)**

(b) Two lines $x - 2y = 5$ and $y - 2x = 5$ makes an angle θ between them. Find θ. **(2)**

8. (a)

$x - y = 8 \quad y = x - 8$

Slope = 1

8. (b)

For $x - 2y = 5$, $m_1 = 1/2$

For $y - 2x = 5$, $m_2 = 2$

$\tan \theta = |(m_2 - m_1)/(1 + m_1 m_2)|$

$= |(2 - 1/2)/(1 + 1)| = 3/4$

$\theta = \tan^{-1}(3/4)$

Answer any 6 questions from 9 to 16. Each carries 4 scores. **(6 × 4 = 24)**

9. (a) If $f(x) = 2x^2 - 5$ and $g(x) = 2x$, then find $(fog)(0)$. **(1)**

(b) Find the range of $f(x) = x^2 + 2; x \in \mathbb{R}$. **(3)**

9. (a)

$f(x) = 2x^2 - 5, \quad g(x) = 2x$

(f composed with g)(0) = 0

9. (b)

$f(x) = x^2 + 2$, x is a real number

Since $x^2 \geq 0$, $x^2 + 2 \geq 2$

Minimum value = 2 at x = 0

Range = [2, infinity)

10. (a) How many relations can be defined from the set A = {x, y, z} to set B = {1, 2} ? **(1)**

(b) Consider the real valued function $f(x) = \begin{cases} 1 - x, & x < 0 \\ 1, & x = 0 \\ x + 1, & x > 0 \end{cases}$

(i) Draw the graph of f(x). **(1)**

(ii) Write the domain and range of f(x). **(2)**

10. (a)

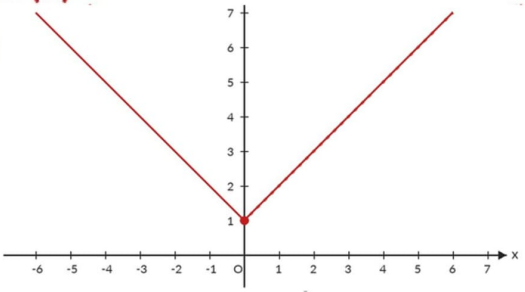
A = {x, y, z} has 3 elements and B = {1, 2} has 2 elements.

Number of elements in A × B = 3 × 2 = 6.

Number of relations = 2^6 = 64.

10. (b)

(i)



(ii) Domain = R Range = [1, infinity)

11. (a) Find the value of i^{102} and $\frac{1}{i^{25}}$. **(2)**

(b) Write the modulus of the complex number $\frac{1+i}{1-i}$. **(2)**

11. (a)

$i^{102} = i^{(4 \times 25+2)} = (i^4)^{25} \times i^2 = 1 \times (-1) = -1$

$1 / i^{25} = 1 / (i^{24} \times i) = 1 / i = -i$

(b)

$z = (1 + i) / (1 - i)$

Multiply numerator and denominator by $(1 + i)$:

$z = (1 + i)^2 / (1 - i)(1 + i) = 2i / 2 = i$

Modulus: $|z| = |i| = 1$

12. How many ways can the letters of the word **PERMUTATIONS** be arranged so that

(a) words start with **P** and end with **S** **(2)**

(b) vowels come together **(2)**

12. (a)

PERMUTATIONS has 12 letters, with T repeated twice.
Fix P at the beginning and S at the end.
Remaining 10 letters contain T twice.
Number of arrangements = 10! / 2!
= 3,628,800 / 2
= 1,814,400

(b)

Vowels: E, U, A, I, O (5 vowels).
Keep all 5 vowels together as one block.
Now there are 8 units, with T repeated twice.
Total number of arrangements = (8! / 2!) x 5!

13. (a) Find the equation of a line passing through the point (1, 4) and (4, 10). **(2)**

(b) Write its intercepts on axes. **(1)**

(c) Find the distance of this line from the origin. **(1)**

(a) Using $y - y_1 = m(x - x_1)$:

$y - 4 = 2(x - 1)$

$y - 4 = 2x - 2$

$\therefore\ y = 2x + 2$

(b) $0 = 2x + 2 \Rightarrow x = -1$

x-intercept = $(-1, 0)$

y-intercept: Put $x = 0$

$y = 2(0) + 2 = 2$

y-intercept = $(0, 2)$

c)

Distance $= |Ax_1 + By_1 + C| / \sqrt{(A^2 + B^2)}$

$= |2(0) - 0 + 2| / \sqrt{(2^2 + (-1)^2)}$

$= 2 / \sqrt{5}$

14. An ellipse passing through (6, 5) having foci $(\pm 6, 0)$. Find its

(a) length of major and minor axis **(2)**

(b) equation **(1)**

(c) eccentricity **(1)**

Given: c = 6 and the ellipse passes through (6, 5).

$x^2/a^2 + y^2/b^2 = 1,\quad a^2 - b^2 = 36$

$36/a^2 + 25/b^2 = 1 \Rightarrow a^2 = 81,\ b^2 = 45$

(a) Major axis $= 2a = 18$

Minor axis $= 2b = 6\sqrt{5}$

(b) Equation: $x^2/81 + y^2/45 = 1$

(c) Eccentricity: $e = c/a = 6/9 = 2/3$

15. (a) Write the number of terms in the expansion of $(2x - y)^{99}$. **(1)**

(b) Using Binomial theorem, show that $6^n - 5n$ always leaves remainder 1 when divided by 25. **(3)**

15. (a)

Number of terms in $(2x - y)^{99}$ =

99 + 1 = 100

(b)

$6^n = (1 + 5)^n$

By Binomial theorem,

$6^n = 1 + 5n +$ terms containing 25

$6^n - 5n = 1 + 25k$, where k is an integer

Therefore, the remainder is 1 when divided by 25.

16. (a) If A is an event of a random experiment with $P(A) = \frac{3}{5}$, then find $P(\overline{A})$. **(1)**

(b) If a fair coin with 1 marked on one face and 6 on the other face and a fair die are tossed together, then find the probability that the sum of the numbers turn up is 12. **(3)**

16. (a)

$P(A) = 3/5$

$P(\overline{A}) = 1 - P(A) = 1 - 3/5 = 2/5$

(b)

Possible outcomes: (1,1), (1,2), ..., (1,6),
(6,1), (6,2), ..., (6,6)

Sum = 12 only for (6,6).

P(sum = 12) = 1/12

Answer any 3 questions from 17 to 20. Each carries 6 scores. **(3 × 6 = 18)**

17. (a) Evaluate $\sec\left(\frac{13\pi}{6}\right)$. **(1)**

(b) Prove that $\frac{\tan\left(\frac{\pi}{4}+x\right)}{\tan\left(\frac{\pi}{4}-x\right)}=\left(\frac{1+\tan x}{1-\tan x}\right)^2$. **(2)**

(c) Prove that $(\cos x-\cos y)^2+(\sin x-\sin y)^2=4\sin^2\left(\frac{x-y}{2}\right)$. **(3)**

17. (a)

$\sec(13\pi/6) = \sec(\pi/6) = 2/\sqrt{3}$

(b) Let t = tan x.

$\tan(\pi/4 + x) = (1 + t)/(1 - t)$

$\tan(\pi/4 - x) = (1 - t)/(1 + t)$

Therefore,

$\tan(\pi/4 + x) / \tan(\pi/4 - x)$

$= [(1 + t)/(1 - t)]^2$

$= [(1 + \tan x)/(1 - \tan x)]^2$

(c)

$(\cos x - \cos y)^2 + (\sin x - \sin y)^2$

$= 2 - 2\cos(x - y)$

$= 4\sin^2((x - y)/2)$

Hence proved.

18. (a) Write the first three terms of the sequence formed by $a_n = \frac{n}{n+1}$; $n \in \mathbb{N}$, natural number. **(2)**

(b) The fourth term of a geometric progression is square of its second term and the first term is –3. Find its seventh term. **(2)**

(c) Find the sum of first 20 terms of the geometric progression $\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \ldots$ **(2)**

(a) $a_n = n/(n + 1)$

$a_1 = 1/2, \quad a_2 = 2/3, \quad a_3 = 3/4$

(b)

First term $a = -3$

Let common ratio be r.

Fourth term = ar^3, second term = ar

$ar^3 = (ar)^2 \Rightarrow -3r^3 = 9r^2$

$r = -3$

Seventh term = $ar^6 = -3(-3)^6 = -2187$

c) GP: $\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \ldots$

$a = \sqrt{7}, \quad r = \sqrt{3}$

$S_{20} = a(r^{20} - 1)/(r - 1)$

$= \sqrt{7}(3^{10} - 1)/(\sqrt{3} - 1)$

19. (a) Find the derivative of $\cos x$ using first principle. **(2)**

(b) Find the derivative of

(i) $\dfrac{x^2}{x+1}$ **(2)**

(ii) $\sin x \cos x$ **(2)**

(a) $f'(x) = \lim_{h\to 0} [\cos(x + h) - \cos x]/h$

$= \lim_{h\to 0} [\cos x(\cos h - 1) - \sin x \sin h]/h$

$= -\sin x$

(b)

$d/dx [x^2/(x + 1)]$

$= [2x(x + 1) - x^2]/(x + 1)^2$

$= x(x + 2)/(x + 1)^2$

(ii)

$d/dx (\sin x \cos x)$

$= \cos^2 x - \sin^2 x$

$= \cos 2x$

20. (a) Find Mean Deviation of the following data :

3, 9, 5, 3, 12, 10 **(2)**

(b) Find the Standard Deviation of the following observations :

x_i	6	10	14	18	24	28	30
f_i	2	4	7	12	8	4	3

(4)

20. (a)

Data: 3, 9, 5, 3, 12, 10

Mean = 42/6 = 7

Mean deviation = $\Sigma|x - \bar{x}|/n$

= (4 + 2 + 2 + 4 + 5 + 3)/6 = 10/3

(b)

N = Σf = 40, Σfx = 760

Mean $\bar{x}$ = Σfx/Σf = 760/40 = 19

$\Sigma fx^2 = 16280$

Standard deviation = $\sqrt{[(\Sigma fx^2/N) - \bar{x}^2]}$

= $\sqrt{[(16280/40) - 19^2]}$

= $\sqrt{43.4} \approx 6.59$